

Accelerated Article Preview

Stereoelectronic manipulation of ligands for perovskite solar cells

Received: 24 July 2025

Accepted: 6 May 2026

Accelerated Article Preview

Published online: 13 May 2026

Cite this article as: Yang, T. et al.
Stereoelectronic manipulation of ligands
for perovskite solar cells. *Nature* <https://doi.org/10.1038/s41586-026-10626-0> (2026)

Tinghuan Yang, Erxin Zhao, Nan Wu, Xiaoming Chang, Chenqing Tian, Hai-Long Wang, Lu Zhang, Nannan Gu, Ting Nie, Ye Yang, Zheng Zhang, Tianfei Xu, Xin Chen, Shuang Wang, Tianqi Niu, Niansheng Xu, Chuang Ma, Haojin Li, Buyi Yan, Zicheng Ding, Shengzhong Frank Liu, Feng Gao & Kui Zhao

This is a PDF file of a peer-reviewed paper that has been accepted for publication. Although unedited, the content has been subjected to preliminary formatting. Nature is providing this early version of the typeset paper as a service to our authors and readers. The text and figures will undergo copyediting and a proof review before the paper is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers apply.

Stereoelectronic manipulation of ligands for perovskite solar cells

Tinghuan Yang,¹ Erxin Zhao,¹ Nan Wu,¹ Xiaoming Chang,¹ Chenqing Tian,¹ Hai-Long Wang,² Lu Zhang,³ Nannan Gu,³ Ting Nie,¹ Ye Yang,¹ Zheng Zhang,¹ Tianfei Xu,¹ Xin Chen,¹ Shuang Wang,¹ Tianqi Niu,¹ Niansheng Xu,⁴ Chuang Ma,¹ Haojin Li,¹ Buyi Yan,³ Zicheng Ding,¹ Shengzhong (Frank) Liu,^{1,5,*} Feng Gao,^{4,*} Kui Zhao^{1,*}

1 Key Laboratory of Applied Surface and Colloid Chemistry, Ministry of Education; Shaanxi Key Laboratory for Advanced Energy Devices; Shaanxi Engineering Lab for Advanced Energy Technology; Institute for Advanced Energy Materials; School of Materials Science and Engineering, Shaanxi Normal University, Xi'an 710119, China.

2 State Key Laboratory of Physical Chemistry of Solid Surfaces, College of Chemistry and Chemical Engineering, Innovation Laboratory for Sciences and Technologies of Energy Materials of Fujian Province (IKKEM), Xiamen University, Xiamen 361005, China.

3 Hangzhou Microquanta Semiconductor Co., Ltd., Hangzhou 310030, China.

4 Department of Physics, Chemistry and Biology (IFM), Linköping University, Linköping, Sweden.

5 Key Laboratory of Photoelectric Conversion and Utilization of Solar Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, 116023, Liaoning, China.

*Corresponding author Email: zhaok@snnu.edu.cn (K.Z.), feng.gao@liu.se (F.G.), szliu@dicp.ac.cn (S.L.)

Abstract

Interfacial losses at perovskite/charge transport layer heterojunctions persist as a critical barrier to achieving high-performance perovskite solar cells.¹⁻⁵ While molecular ligands can passivate interfacial vacancy defects, their vertical anchoring geometry compromises charge transport by increasing interfacial transport pathway. Here, we demonstrate that stereoelectronic manipulation of ligand adsorption topology advances interfacial minimum energy loss for efficient and stable perovskite solar cells. By strategically replacing benzene carbons with nitrogen atoms to create pyridine or pyrimidine rings, we design ligands that concurrently anchor to the perovskite through Pb-N coordination bonds and Pb-I- π interactions, endowing a single molecule with dual, synergistic binding modes. This mutually reinforcing stereoelectronic interplay drives thermodynamically favorable planar alignment of ligands, enabling atomic-scale defect mitigation while maintaining sub-nanometer-scale charge transfer across the interface. The optimized interfacial architecture achieves a stabilized power output of 26.85%, with certificated reverse-scan and forward-scan efficiencies of 27.41% and 26.35%, respectively. Furthermore, the solar modules exhibit exceptional operational stability, retaining 85.8% of initial module efficiency after 258 days of outdoor real-time field testing.

Main

Perovskite solar cells (PSCs) have redefined photovoltaic technology frontiers through unprecedented efficiency gains, achieving certified power conversion efficiencies (PCEs) approaching 27% in laboratory settings.⁶⁻⁹ However, there remain losses at the perovskite/charge transport layer (CTL) interfaces - a critical barrier to both efficiency ceilings and industrial scalability.¹⁻⁵ The losses at perovskite/CTL interfaces are fundamentally governed by two quantum mechanically antagonistic processes.¹⁰⁻¹³ One is defect-mediated non-radiative recombination, which is dominated by Shockley-Read-Hall (SRH) kinetics with characteristic abundant interfacial trap states at untreated interfaces.¹⁴⁻¹⁶ Another is interfacial transport resistance limited by quantum-mechanical transmission and wavefunction mismatch across the heterojunction.^{17,18}

Existing molecular design strategies face an intrinsic trade-off. Ligands (e.g., phenethylammonium) achieve defect passivation via intercalation of amine terminal into the Pb-I framework for ionic interaction bonding, but impose prohibitive transport distance ($d > 2$ nm) and parasitic transport resistance due to outside carbon-chain organic moieties.¹⁹⁻²⁴ This trade-off poses a fundamental challenge in molecular design: atomic-scale defect mitigation requires strong interfacial bonding (vertical alignment), while optimal charge transport demands minimized transport distance (planar alignment). Resolving this trade-off requires precise modulation of interfacial energy losses through synergistic defect passivation and charge transport geometry optimization, thereby enabling minimum interfacial losses. Yet, even state-of-the-art remedies including those employing vertical ligands or geometric size-matching passivants,^{13,24-26} leave this fundamental dilemma unresolved.

Herein, we report stereoelectronic manipulation of ligands that simultaneously suppresses atomic-scale defects and optimizes molecule-level charge transport at perovskite interfaces. By integrating N-heteroaromatic motifs into ligands, we reorient their adsorption topology from vertical to planar configurations via dual Pb-N coordination and Pb-I- π interactions with the perovskite lattice. This interfacial reconfiguration enhances interfacial binding energy, reduces charge transport distance, and suppresses energy disorder, while accelerating charge extraction. Through

synergistic energy loss management, our devices achieve a certified stabilized power output of 26.85%. Our ligand design strategy is also applicable to wide-gap perovskites and large-area perovskite solar cells/modules. Moreover, large-area modules retain 85.8% of initial efficiency (23.10%) after 258 days outdoor real-time field testing, demonstrating unprecedented efficiency-stability synergy.

Stereoelectronic manipulation of ligands

The stereoelectronic design principle was designed to precisely control ligand adsorption topology through nitrogen-mediated orbital hybridization. Four multifunctional ligands were engineered with amidino (FA^+ , for the Pb-N ionic bonding) as defect passivation and $-\text{CF}_3$ (hydrophobic moiety) as enhancement of the environmental stability for the perovskite surface.^{22,27-29} At the center of the ligands, we introduced conjugated benzene rings and progressive nitrogen incorporation in aromatic cores gradually replaced the C on the benzene rings with N to form N-heteroaromatic motifs, thereby modulating the interaction ability between the central part of the ligands and Pb-I framework. **Fig. 1a** shows the four amidino-based derivatives: trifluoroethylamidine (TFEA), trifluorobenzamidine (TFBA), trifluoropyridylamidine (TFPA), and trifluoropyrimidylamidine (TFPmA), distinguished by their central conjugated cores (highlighted in red) - benzene, pyridine (1 N atom), and pyrimidine rings (2 N atoms), respectively.

First-principles density functional theory (DFT) calculations were used to elucidate the stereoelectronic factors governing ligand adsorption topology on the Pb-I framework of perovskites (See details in **Supplementary Note 1**). Electrostatic potential (ESP) maps revealed increasingly negative surface regions associated with N-heteroaromatic motifs (**Supplementary Fig. 1**), concomitant with a rise in molecular dipole moment from 4.18 D for TFEA to 4.38 D, 6.39 D, and 6.48 D for TFBA, TFPA, and TFPmA, respectively (**Supplementary Fig. 2**). We evaluated adsorption energies (E_{ads}) for each ligand in both vertical and planar orientations on the Pb-I framework. Vertically adsorbed E_{ads} values were -0.97 eV (TFEA), -1.16 eV (TFBA), -1.13 eV (TFPA) and -1.10 eV (TFPmA), whereas planar adsorption energies were -0.40 eV, –

1.15 eV, −1.31 eV and −1.37 eV, respectively (**Fig. 1b**, **Supplementary Fig. 3**). The difference $\Delta E = E_{\text{ads}}(\text{planar}) - E_{\text{ads}}(\text{vertical})$ decreased from 0.57 eV (TFEA) to almost zero (0.01 eV) for TFBA and became negative for TFPA (−0.18 eV) and TFPmA (−0.27 eV), indicating a thermodynamic preference shift from the vertical (TFEA) to random (TFBA) and then planar (TFPA, TFPmA) topology as the number of N-heteroaromatic units increases (**Fig. 1c**). This thermodynamic preference shift was also observed on FA-I-terminated surfaces (**Supplementary Figs. 4 and 5**, and **Supplementary Note 1**). Charge-density difference analysis (**Fig. 1d**) indicated that the planar topology enabled multicenter interactions—namely, Pb–N coordination bonds and Pb–I– π interactions between Pb–I frameworks and ligands.

We next examined how ligand adsorption topology modulates charge transport across the perovskite/C₆₀ interface. Using TFBA in a vertical topology and TFPmA in a planar topology as representative cases, we found that the vertical transport distance (d_{vertical}) from the perovskite surface to the C₆₀ acceptor was 10.60 Å, whereas the corresponding distance for the planar topology (d_{planar}) contracted by ~37 % to 6.64 Å (**Fig. 1e**). Upon reorientation from vertical to planar, the N-heteroaromatic rings on the ligand approached the C₆₀ molecule more closely, resulting in enhanced perovskite/ligands/C₆₀ interactions. Consequently, the overall binding energy of the perovskite/ligands/C₆₀ interface decreased from −0.44 eV for TFBA to −0.59 eV for TFPmA. Theoretical simulation reveals that the ligand adsorption topology opens both sub-nanometer charge-transfer channels and enhances the perovskite-ligand-C₆₀ electrostatic coupling, yielding an interfacial electronic landscape that favours efficient charge extraction.

Multicenter interactions resolved via spectroscopies

To elucidate the chemical mechanisms of stereoelectronic factors governing multicenter interactions between ligands and Pb-I framework, we conducted X-ray photoelectron spectroscopy (XPS), ¹H nuclear magnetic resonance (¹H-NMR), and Fourier-transform infrared spectroscopy (FTIR) for series of samples (see details in **Supplementary Note 2**). The XPS spectra of calibrated C 1s and F 1s references for FAPbI₃-based perovskite with ligands are provided in **Supplementary Figs. 6 and 7**.

As shown in **Figs. 2a–c**, monotonic binding energy shifts were observed in Pb 4f_{7/2} (138.49 → 138.84 eV), I 3d_{3/2} (631.09 → 630.63 eV), and N 1s (400.81 → 400.62 eV) of the perovskite/ligand systems with increasing N-heteroaromatic content. These results indicate that N-heteroaromatic rings enhanced charge redistribution of perovskite with Pb–N coordination bonds and Pb–I– π interactions between Pb–I frameworks and ligands.

¹H NMR spectra indicated a paramagnetic shifting (7.47→10.0 ppm) of the amidino proton δH_α with progressive N-heteroaromatics of four ligands (**Fig. 2d**, **Supplementary Figs. 8–11**), accompanied by peak splitting ($\Delta\delta = 0.27$ ppm for TFPmA). Likewise, the conjugated ring δH_β exhibited deshielding (8.02→9.49 ppm, $\Delta\delta = 1.47$ ppm) with progressive N-heteroaromatics. Upon PbI₂ interaction, δH_α moved to high field for all ligands (blue-red dashed line), suggesting the strong NH...I hydrogen-bonding interactions between the Pb–I framework and the ligand amidino moiety.^{30,31} The distance between δH_α and δH_β gradually decreased and eventually overlapped ($\Delta\delta$: 0.94→0.12→–0.04 ppm, red shaded region) with progressive N-heteroaromatics, reflecting chelate-assisted Pb–I– π delocalization and a transition from monodentate to multicenter interactions.

FTIR spectroscopy of ligands/PbI₂ complexes (**Fig. 2e** and **Supplementary Figs. 12 and 13**) further confirmed multicenter interactions underpinning geometric transitions. Upon PbI₂ complexation, the amidino C=N stretch in TFEA upshifted by +12 cm^{–1} (1,580→1,592 cm^{–1}), confirming dominant Pb–N coordination interaction. Introducing N-heteroaromatic motifs reversed this trend: TFBA and TFPA exhibited C=N downshifts ($\Delta\nu = -4/-17$ cm^{–1}) with full width at half-maximum broadening (FWHM: 19→29/30→48 cm^{–1}) (**Fig. 2f**). TFPmA further amplified this effect, displaying strong broadening by 107% (FWHM: 36→75 cm^{–1}) of C=N stretch and a –3 cm^{–1} redshift in the C–N vibration (1,354→1,351 cm^{–1}). Concurrently, N-heteroaromatics participation emerged via split vibrational modes (1,590–1,400 cm^{–1}): pyridine N-Pb interaction softened ring breathing (1,571/1,526 cm^{–1}), while pyrimidine distortion split the symmetric stretch into a 1,550/1,502/1,452 cm^{–1} triple (**Fig. 2e**, shaded area lines 1–3), which demonstrate direct engagement of N-heteroaromatic motifs in Pb–I– π interaction.

The spectroscopic observations align with DFT-predicted multicenter interactions between ligands and Pb-I framework, validating stereoelectronic engineering as a tool for atomic-scale control of ligand orientation and interaction cooperativity.

Characterizations of perovskite/ligands interface

Complementary grazing-incidence wide-angle X-ray scattering (GIWAXS), X-ray diffraction (XRD) and ultrafast transient absorption spectroscopy (TAS) analyses of FAPbI₃-based perovskite films verified that interfacial treatment preserved the intrinsic crystallographic integrity of perovskite phases, with no detectable low-dimensional phase formation or lattice distortion (**Supplementary Figs. 14-16**). This confirms molecular-level ligand anchoring on perovskite surface rather than bulk intercalation. Scanning electron microscopy (SEM) revealed dissipation of surface-aggregated PbI₂ domains after interfacial treatment (**Supplementary Fig. 17**), yielding atomically smooth interfaces with ~1.4–1.7 nm reduced root-mean-square roughness (**Supplementary Fig. 18**). Kelvin probe force microscopy (KPFM) quantified a ~61.8% decrease in surface potential differential (30.1 → 11.5 mV, **Fig. 3a**) with significantly narrowed spatial heterogeneity ($\pm 150 \rightarrow \pm 50$ mV, **Fig. 3b**, **Supplementary Fig. 19**) with progressive N-heteroaromatic motifs. The improved surface smoothness and potential homogenization, driven by the N-heteroaromatic motifs, enhance the contact-related interfacial quality and is beneficial for interfacial charge extraction.

Hyperspectral nano-infrared imaging (nano-IR; <10 nm resolution, **Fig. 3c**, and **Supplementary Note 3**) mapped nanoscale spatial heterogeneity of ligands on perovskite surfaces.³²⁻³⁵ Untreated films showed uniform FA cations ($\text{-C}\equiv\text{N}$, 1712 cm⁻¹) with PbI₂-rich clusters (**Supplementary Fig. 20**). Ligand-treated surfaces retained this signature but exhibited distinct vibrational bands (**Fig. 3d**, **Supplementary Fig. 21**): C-N bending (TFEA, 1396 cm⁻¹), C $\equiv$ N/C-N stretches (TFBA: 1674/1325 cm⁻¹; TFPA: 1682/1337 cm⁻¹; TFPmA: 1682/1352 cm⁻¹), corroborated by FTIR (**Fig. 2d**). Multivariate analysis (**Fig. 3c**, **Supplementary Fig. 22**, and **Supplementary Note 3**) resolved four relatively signal-active adsorption modes: sparse fractal aggregates (TFEA, ~7.7% relative signal-active fractions), discontinuous superstructures (TFBA,

~25.7%), aligned nanowires (TFPA, ~33.6%), and near-complete epitaxial nanodomains (TFPmA, ~46.3%). TFPmA achieved intra-grain uniformity, reducing average aggregation size by 39% versus TFEA (~39 vs. ~100 nm, **Supplementary Table 1**), establishing a conformal passivation network that suppresses interfacial traps.

Ultraviolet photoelectron spectroscopy (UPS) revealed a systematic modulation of interfacial energy landscapes governed by ligand stereoelectronic effects. Work functions decreased from 4.85 eV (control) to 4.72 eV (TFPmA) (**Supplementary Fig. 23**, and **Supplementary Table 2**), correlating with charge redistribution by the strong surface interaction/dipole of N-heteroaromatic motifs.^{16,36} This trend reflects flattened surface energy undulation, consistent with the reduced spatial potential fluctuations observed by KPFM, which mitigates interfacial charge recombination through reduced electrostatic disorder. Concomitantly, the perovskite conduction band minimum (CBM), which was derived from the UPS-determined valence band maximum (VBM) and the optical bandgap (**Supplementary Figs. 23-25**, and **Supplementary Table 2**),^{22,37} shifted downward from -4.11 eV (control) to -4.21 eV (TFPmA) (**Fig. 3e**), thereby narrowing the energy offset with C₆₀ (-4.24 eV) from 0.13 to 0.03 eV. Molecular orbital calculations and differential charge-density maps indicated that ligand-perovskite multicenter interactions induced interfacial charge redistribution, which in turn promoted interfacial energy-level realignment (**Supplementary Figs. 26 and 27**, and **Supplementary Table 3**). The sub-0.1 eV energy offset at the perovskite/TFPmA/C₆₀ interface drives the energy-barrier to the theoretical minimum, enabling quasi-ideal charge extraction across the heterojunction.

Quasi-Fermi level splitting (QFLS) analysis for FTO/4PADCB/perovskite/ligands samples quantified a hierarchy of perovskite/ligands interfacial defect suppression, showing a monotonic increase from 1.175±0.012 eV (control, without ligands) to 1.236±0.009 eV (TFPmA) (**Fig. 3f**). This 61 meV enhancement, enabled by stereoelectronic manipulation, stems from above observed synergistic interfacial mechanisms: topographic defect healing through multicenter interactions via Pb-N coordination, delocalized electronic screening mediated by Pb-I- π interactions, a

reduction in the interfacial energetic barrier,^{13,24} and a rise in molecular dipole moment that reshapes the local electrostatic landscape.^{38,39}

Ligands anchoring and charge dynamics

The ligands anchoring and charges dynamics at the perovskite/CTL interface during solution-processing were investigated using *in situ* photoluminescence (PL) spectroscopy with a 405 nm laser (see details in **Experimental section**). We spin-coated the ligand solution onto the FAPbI₃-based perovskite film and then drop-cast a solution of [6,6]-phenyl-C₆₁-butyric acid methyl ester (PCBM) instead of insoluble C₆₀ to form the CTL (**Supplementary Fig. 28**). **Fig. 4a** shows two-dimensional PL intensity evolution maps recorded with and without surface ligands, and **Fig. 4b** traces the evolution of the 788 nm emission peak over time (extracted PL spectra at 5, 25, 42, and 60 s are given in **Supplementary Figs. 29-32**).

Upon contact with the ligand solution, the PL intensity displayed a two-stage anchoring process. In the initial rapid stage ($< \sim 3$ s), PL increased by $\sim 18\%$ relative to the neat perovskite (control), reflecting immediate ligand adsorbing at the perovskite surface in solution. A subsequent, slower stage (~ 3 – 20 s) driven by solvent evaporation then ensued. The PL intensity further increased (up to 28% for TFPmA; **Fig. 4c**), consistent with ongoing defect passivation that may involve Pb–N coordination and Pb–I– π interactions.^{12,40}

Upon PCBM addition, the PL intensity exhibited a biphasic decay, with an initial rapid quenching phase (at ~ 42.5 – 45 s) from maximum value ($I'_{\max}$) followed by a slower decrease until stabilization (> 45 s). The sub-2.5 s quenching arises from ultrafast electron transfer from perovskite to CTL during film deposition, even in the presence of residual solvent. This kinetic profile terminated upon complete solvent evaporation and PCBM film solidification, marking the conclusion of interfacial charge-extraction dynamics. Notably, TFPmA-modified films retained only 35 % of their maximum PL intensity—versus 78% retention in the unmodified control (**Fig. 4d**)—indicating enhanced charge extraction efficiency at the perovskite/ligand/CTL interface with a planar adsorption topology. This experimental outcome corroborated DFT predictions

(**Fig. 1e**) that planar adsorption topology increases electrostatic coupling and shortens charge transport distance across the perovskite/ligands/CTL interface, thereby facilitating more efficient charge extraction.

Time-resolved photoluminescence (TRPL) revealed the dual role of ligand adsorption topology in simultaneously passivating defects and enabling ultrafast charge extraction. In perovskite/ligands architectures, TFPmA-modified films exhibited a 195% prolonged average carrier lifetime (3.19 μ s vs. control 1.08 μ s, **Fig. 4e, Supplementary Table 4**). This defect suppression hierarchy—scaling with N-heteroaromatic motifs—arises from multicenter interactions locking undercoordinated Pb^{2+} sites and I^- vacancies. Strikingly, C_{60} integration inverted this trend: TFPmA-treated devices showed an 83% shortened carrier lifetime (11.0 ns vs. control 65.9 ns, **Fig. 4f, Supplementary Table 5**), indicative of improved charge extraction efficiency enabled by adsorption topology induced interfacial dipole alignment. The lifetime reduction efficacy followed the order of TFPmA > TFPA > TFBA > TFEA (16.7% $\rightarrow$ 69.0%), mirroring DFT-predicted charge transport distance compression by $\sim 37\%$. Spatially resolved TRPL mapping provided complementary insights. The ligands-treated films displayed homogeneous distribution than the control film for the TFEA, TFBA, TFPA and TFPmA (**Supplementary Figs. 33, 34**), respectively, verifying increased spatially consistence which aligns with nano-IR observation (**Fig. 3d**). Collectively, this duality—lifetime extension in passivated systems versus acceleration in extraction-optimized interfaces—demonstrates efficient interfacial carrier management through stereoelectronic ligand regulation.

Mott-Schottky analysis quantified enhanced built-in potential (V_{bi}) from 1.098 V (control) to 1.199 V (TFPmA) (**Supplementary Fig. 35**), in line with the improvement in charge-separation efficiency suggested by the QFLS data. Electrochemical impedance spectroscopy (EIS) revealed ligand-dependent interface optimization: series resistance (R_s) decreased by 78% (65.8 $\rightarrow$ 14.4 Ω) while recombination impedance (R_{rec}) surged 8.9-fold (3,314 $\rightarrow$ 29,690 Ω) (**Supplementary Fig. 36, and Supplementary Table 6**), demonstrating suppression of interfacial recombination which stems from N-heteroaromatic motif contribution including contact-related morphological

optimization and multicenter coordination-induced electronic microenvironment enhancement.^{40,41}

Device performance and stability

Leveraging stereoelectronic ligand manipulation, we fabricated p-i-n perovskite solar cells (FTO/4PADCB/Perovskite (FAPbI₃-based)/ligands/C₆₀/BCP/Ag, 0.074 cm² cell, **Fig. 5a**), achieving PCEs of 24.26%, 24.65%, 25.98%, 26.37% and 27.58% for devices without (control) and with TFEA, TFBA, TFPA and TFPmA modifications (**Fig. 5b**), respectively. Owing to the presence of hysteresis, all efficiency parameters are extracted from reverse-scan *J*-*V* curves unless otherwise stated. Ligand-modified devices exhibited progressively reduced hysteresis indices (5.9%→2.1%) (**Supplementary Fig. 37**, and **Supplementary Table 7**). TFPmA-based cells delivered $V_{oc} = 1.218$ V, $J_{sc} = 26.13$ mA cm⁻², and FF = 86.69%, with a hysteresis index of 2.1%. We sent the best-performing device to an accredited PV laboratory (the PV Metrology Lab of National Institute of Metrology, China) for third-party certification and obtained a reverse-scan efficiency of 27.41% with stabilized power output efficiency of 26.85% (**Fig. 5c**, **Supplementary Figs. 38-41**). The external quantum efficiency (EQE) integrated currents agree with *J*-*V* measurements (tolerance variation < 3.2%) and the calculated bandgap of 1.556 eV confirmed a voltage loss of only 0.338 V (**Supplementary Fig. 42**). Statistical analysis of 20 independent devices confirmed excellent reproducibility of the ligand modification strategy (**Supplementary Fig. 43**, and **Supplementary Table 8**). The strategy is readily applicable to wide-bandgap perovskite systems (1.68 eV), yielding 24.04% PCE (certified at 23.04%, **Fig. 5d**, **Supplementary Figs. 44** and **45**, and **Supplementary Table 9**). Unencapsulated devices retained 90% initial PCE after 1,056 h at 85°C (ISOS-D-2) and 2,190 h MPP tracking (ISOS-L-1) (**Supplementary Figs. 46** and **47**), delivering excellent operational stability.

Building on stereoelectronic manipulation, we demonstrated scalable device fabrication using TFPmA interfacial modification. In 1.098 cm² single-junction cells, we achieved a champion PCE of 26.40% (certified at 25.62%, **Fig. 5e**, **Supplementary**

Figs. 48 and 49) with $V_{OC} = 1.202$ V, $J_{SC} = 25.54$ mA cm⁻², FF = 86.04%. Extending the active area to 13.95 cm², modules achieved a champion PCE of 24.74% ($V_{OC} = 7.160$ V, $J_{SC} = 4.18$ mA cm⁻², FF = 82.73%, **Fig. 5f, Supplementary Table 10**). These results underscore the compatibility of planar-adsorbed ligands with scalable fabrication and their ability to preserve uniform, high-performance interfaces over large areas.

We performed real-time field testing of our large-area perovskite modules under actual out-door operating conditions to investigate interfacial stability (**Fig. 5g, Supplementary Figs. 50 and 51**). Subjected to 258 days of continuous operation with temperature cycling (-11.5 to 68.9 °C) and irradiance fluctuations (0–1.39 sun; plane of array global irradiance (PGD) of 0–34.7 mW cm⁻²), the TFPmA-treated module exhibited only a 14.2% loss (from 23.10% to 19.82% PCEs). In sharp contrast, the control module degraded by 24.6% (from 21.03% to 15.85% PCEs) after merely 27 days of outdoor testing (**Supplementary Fig. 52**). The results from ISOS-D-2 testing, ISOS-L-1 testing, and real-time outdoor field testing collectively demonstrate that stereoelectronic manipulation of ligands not only minimizes energy-loss pathways at the perovskite/CTL interface but also endows devices with remarkable operation robustness at practical operating conditions, which would provide a compelling technical foundation for reliable large-scale perovskite photovoltaics deployment.

Conclusions

This work demonstrates that the ligand adsorption topology on perovskite surfaces, governed by ligand-perovskite interactions, enables minimization of interfacial energy losses. Introducing N-heteroaromatic motifs into ligands triggers a stereoelectronic reconfiguration from vertical to planar alignment via synergistic Pb-N coordination and Pb-I- π interactions. This topology change concurrently annihilates defects while sustaining sub-nanometer charge transport channels, achieving a stabilized power output of 26.85%. Notably, the strategy is both bandgap-universal and readily scalable, delivering high efficiencies for wide-bandgap (1.68 eV) devices and large-area cells and modules. Extended outdoor validation reveals an interfacial robustness during 258

days real-time field testing. This atomic-scale interface control paradigm provides new design principles for minimizing interfacial energy losses to advance perovskite photovoltaics toward their theoretical efficiency limit while ensuring scalability and durability.

References

- 1 Liu, S. *et al.* Buried interface molecular hybrid for inverted perovskite solar cells. *Nature* **632**, 536–542 (2024).
- 2 Isikgor, F. H. *et al.* Molecular engineering of contact interfaces for high-performance perovskite solar cells. *Nat. Rev. Mater.* **8**, 89–108 (2023).
- 3 Rajagopal, A. *et al.* Toward Perovskite Solar Cell Commercialization: A Perspective and Research Roadmap Based on Interfacial Engineering. *Adv. Mater.* **30**, 1800455 (2018).
- 4 Wang, H. *et al.* Impurity-healing interface engineering for efficient perovskite submodules. *Nature* **634**, 1091–1095 (2024).
- 5 Liang, Q. *et al.* Highly stable perovskite solar cells with 0.30 voltage deficit enabled by a multi-functional asynchronous cross-linking. *Nat. Commun.* **16**, 190 (2025).
- 6 NREL. *National Renewable Energy Laboratory*, Best Research-Cell Efficiencies Chart, <https://www.nrel.gov/pv/cell-efficiency.html>, (2026).
- 7 Dai, Z. *et al.* Interfacial toughening with self-assembled monolayers enhances perovskite solar cell reliability. *Science* **372**, 618–622 (2021).
- 8 Huang, J. *et al.* Understanding the physical properties of hybrid perovskites for photovoltaic applications. *Nat. Rev. Mater.* **2**, 17042 (2017).
- 9 Park, N.-G. *et al.* Towards stable and commercially available perovskite solar cells. *Nat. Energy* **1**, 16152 (2016).
- 10 Lin, Y. *et al.* A Nd@C82-polymer interface for efficient and stable perovskite solar cells. *Nature* **642**, 78–84 (2025).
- 11 You, S. *et al.* C₆₀-based ionic salt electron shuttle for high-performance inverted perovskite solar modules. *Science* **388**, 964–968 (2025).
- 12 Zhu, H. *et al.* In situ energetics modulation enables high-efficiency and stable inverted perovskite solar cells. *Nat. Photon.* **19**, 28–35 (2025).
- 13 Chen, H. *et al.* Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* **384**, 189–193 (2024).
- 14 De quilettes, D. W. *et al.* Impact of microstructure on local carrier lifetime in perovskite solar cells. *Science* **348**, 683–686 (2015).
- 15 Ni, Z. *et al.* Resolving spatial and energetic distributions of trap states in metal halide perovskite solar cells. *Science* **367**, 1352–1358 (2020).
- 16 Xiong, S. *et al.* Reducing nonradiative recombination for highly efficient inverted perovskite solar cells via a synergistic bimolecular interface. *Nat. Commun.* **15**, 5607 (2024).

- 387 17 Laikhtman, B. *et al.* Landauer formula for transmission across an interface. *Phys.*
388 *Rev. B* **49**, 17177–17184 (1994).
- 389 18 Lian, Q. *et al.* Dual molecular bridges at perovskite heterointerfaces for
390 efficient inverted solar cells. *Natl. Sci. Rev.* **12**, nwaf211 (2025).
- 391 19 Yang, Y. *et al.* Amidination of ligands for chemical and field-effect passivation
392 stabilizes perovskite solar cells. *Science* **386**, 898–902 (2024).
- 393 20 Jiang, Q. *et al.* Surface reaction for efficient and stable inverted perovskite
394 solar cells. *Nature* **611**, 278–283 (2022).
- 395 21 Wang, Y. *et al.* Homogenized contact in all-perovskite tandems using tailored 2D
396 perovskite. *Nature* **635**, 867–873 (2024).
- 397 22 Chen, R. *et al.* Reduction of bulk and surface defects in inverted methylammonium-
398 and bromide-free formamidinium perovskite solar cells. *Nat. Energy* **8**, 839–849
399 (2023).
- 400 23 Chen, H. *et al.* Quantum-size-tuned heterostructures enable efficient and stable
401 inverted perovskite solar cells. *Nat. Photon.* **16**, 352–358 (2022).
- 402 24 Chang, X. *et al.* Solvent-dripping modulated 3D/2D heterostructures for high-
403 performance perovskite solar cells. *Nat. Commun.* **16**, 1042 (2025).
- 404 25 Shi, Y. *et al.* Tunable molecular dipole moments and orientations for efficient
405 and stable perovskite solar cells. *Joule* **9** (2025).
- 406 26 Huang, H. *et al.* Preferred Parallel Alignment of Sulfonamide Enables High-
407 performance Inverted Perovskite Solar Cells. *Adv. Mater.* **37**, 2507918 (2025).
- 408 27 Yang, T. *et al.* Amidino-based Dion-Jacobson 2D perovskite for efficient and
409 stable 2D/3D heterostructure perovskite solar cells. *Joule* **7**, 574–586 (2023).
- 410 28 Yang, T. *et al.* One-stone-for-two-birds strategy to attain beyond 25% perovskite
411 solar cells. *Nat. Commun.* **14**, 839 (2023).
- 412 29 Park, S. M. *et al.* Engineering ligand reactivity enables high-temperature
413 operation of stable perovskite solar cells. *Science* **381**, 209–215 (2023).
- 414 30 Wu, Y. *et al.* Stable perovskite solar cells with 25.17% efficiency enabled by
415 improving crystallization and passivating defects synergistically. *Energy*
416 *Environ. Sci.* **15**, 4700–4709 (2022).
- 417 31 Liu, T. *et al.* Spacer Engineering Using Aromatic Formamidinium in 2D/3D Hybrid
418 Perovskites for Highly Efficient Solar Cells. *ACS Nano* **15**, 7811–7820 (2021).
- 419 32 Dong, R. *et al.* High-Gain and Low-Driving-Voltage Photodetectors Based on
420 Organolead Triiodide Perovskites. *Adv. Mater.* **27**, 1912–1918 (2015).
- 421 33 Morsch, S. *et al.* Insights into Epoxy Network Nanostructural Heterogeneity Using
422 AFM-IR. *ACS Appl. Mater. Interfaces* **8**, 959–966 (2016).
- 423 34 Tan, Q. *et al.* Inverted perovskite solar cells using dimethylacridine-based
424 dopants. *Nature* **620**, 545–551 (2023).
- 425 35 Fei, C. *et al.* Strong-bonding hole-transport layers reduce ultraviolet
426 degradation of perovskite solar cells. *Science* **384**, 1126–1134 (2024).
- 427 36 Guan, H. *et al.* Controllable Heavy n-type Behaviours in Inverted Perovskite
428 Solar Cells with Non-Conjugated Passivants. *Angew. Chem. Int. Ed.* **64**, 202418606
429 (2025).
- 430 37 Jiang, X. *et al.* Isomeric diammonium passivation for perovskite-organic tandem
431 solar cells. *Nature* **635**, 860–866 (2024).

- 38 Menzel, D. *et al.* Field Effect Passivation in Perovskite Solar Cells by a LiF Interlayer. *Adv. Energy Mater.* **12**, 2201109 (2022).
- 39 Er-raji, O. *et al.* Electron accumulation across the perovskite layer enhances tandem solar cells with textured silicon. *Science* **390**, eadx1745 (2025).
- 40 Park, J. Y. *et al.* Thickness control of organic semiconductor-incorporated perovskites. *Nat. Chem.* **15**, 1745–1753 (2023).
- 41 Truong, M. A. *et al.* Tripodal Triazatruxene Derivative as a Face-On Oriented Hole-Collecting Monolayer for Efficient and Stable Inverted Perovskite Solar Cells. *J. Am. Chem. Soc.* **145**, 7528–7539 (2023).

Figure legends

Fig. 1| Ligand structures and adsorption topologies on the perovskite surface. (a) Chemical structures of the four ligands. (b) Adsorption energies of each ligand on the perovskite surface in vertical and planar topology. (c) Formation-energy differences between vertical and planar adsorption topology for each ligand on the perovskite surface. (d) Differential charge-density maps for ligands adsorbed in vertical versus planar topology, illustrating regions of charge accumulation and depletion. (e) Differential charge-density maps, binding energies, and charge transport distance of perovskite/ligand/C₆₀ interfaces with TFBA or TFPmA.

Fig. 2| The interaction of ligands with the perovskite framework. XPS spectra of (a) Pb 4f, (b) I 3d and (c) N 1s from perovskite films treated with TFEA, TFBA, TFPA, or TFPmA. (d) ¹H-NMR spectra of different ligands interacting with PbI₂. (e) FTIR spectra of different ligands interacting with PbI₂. (f) Full width at half-maximum (FWHM) of the -C=N vibrational peak extracted from the FTIR spectrum.

Fig. 3| Surface characterization of ligand-modified perovskite films. (a) Kelvin probe force microscopy (KPFM) topography showing surface potential variations for control and perovskite films treated with TFEA, TFBA, TFPA, and TFPmA ligands (scale bar: 400 nm). (b) Corresponding cross-sectional potential profiles extracted from the KPFM images in (a). (c) Nano-infrared (nano-IR) chemical mapping of perovskite films modified with TFEA, TFBA, TFPA, and TFPmA (scale bar: 60 nm), showing spatial distribution of key IR absorptions. (d) Nano-infrared (nano-IR) line-plot spectra

acquired from the perovskite surface mappings (**Supplementary Fig. 21**, the blue, green, and red lines corresponding to the blue, green, and red mapping, respectively), highlighting vibrational signatures of each ligand. (e) Energy-level diagrams comparing the valence- and conduction-band edges of untreated (control) versus ligands-treated perovskite films, derived from UPS measurements and optical absorption data. (f) Quasi-Fermi level splitting (QFLS) values determined for FTO/4PADC/BA/PbI₂/perovskite/ligands samples, indicating changes in photovoltage potential.

Fig. 4| *In situ* passivation and carrier behavior. (a) *In situ* photoluminescence (PL) spectroscopy evolution of the FAPbI₃-based perovskite surface during ligands and the consequent charge transport layer (CTL) spin-coating processes. (b) Normalized PL intensity extracted from *in situ* PL evolution maps. (c) The ratio of the PL intensity before (I_0) and after ligands modification (I_{ligands}). (d) The ratio of the PL intensity before (I'_{max}) and after CTL spin coating (I'_{min}). (e) Time-resolved photoluminescence (TRPL) curves of control and ligands-treated films with glass/perovskite/ligands structure. (f) TRPL curves of control and ligand-treated films with glass/perovskite/ligands/C₆₀ structure.

Fig. 5| Device performance and stability. (a) Cross-sectional structural image of perovskite solar cell (the scale bar is 500 nm). (b) *J-V* curves of the control and ligands treated devices. (c) Certified photovoltaic performance of devices obtained at NIM. (d) *J-V* curves of wide-bandgap (1.68 eV) devices for control and TFPmA treated devices. (e) *J-V* curves of 1.098 cm² devices for TFPmA-treated solar cells. (f) *J-V* curves of the TFPmA-treated module devices. (g) The real-time field testing of TFPmA-treated module under actual out-door operating conditions for 258 days.

Methods

Materials

FAPbI₃ single-crystal powder (99.99%) was purchased from Perov-Opto Co., Ltd. Pb (II) iodide (PbI₂, 99.999%) was purchased from TCI. Methylammonium iodide (MAI, 99.5%) and Methylammonium chloride (MACl, 99.5%) were purchased from Greatcell solar. Cesium iodide (CsI, 99.99%), [6,6]-phenyl-C₆₁-butyric acid methyl ester (PCBM), Pb (II) bromide (PbBr₂, 99.99%) were purchased from Advanced Election Technology Co., Ltd. C₆₀, (4-(7H-dibenzo[c,g]carbazol-7-yl)butyl)phosphonic acid (4PADCB), and Pb (II) chloride (PbCl₂, 99.99%) were purchased from Xi'an Yuri Solar Co., Ltd. 2,2,2-Trifluoroacetimidamide hydrochloride (TFEA, 95%) and 4-(Trifluoromethyl)benzimidamide hydrochloride (TFBA, 97%), were purchased from Bide Pharmatech Co., Ltd. 6-(Trifluoromethyl)nicotinimidamide hydrochloride (TFPA, 99%), 2-(Trifluoromethyl)pyrimidine-5-carboximidamide hydrochloride (TFPmA, 95%) were purchased from Leyan. Chlorobenzene (CB) (anhydrous, 99.8%) and isopropanol (IPA) (anhydrous, 99.5%) was purchased from Sinopharm Chemical Reagent Co., Ltd. N,N-dimethylformamide (DMF) (anhydrous, 99.8%), dimethyl sulfoxide (DMSO) (anhydrous, ≥99.9%), and 2,9-Dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP) were purchased from Alfa Aesar. Nickel acetylacetonate dihydrate (Ni(acac)₂·2H₂O, 99%) were purchased from Shanghai Aladdin. All the chemical materials and reagents were used as received without further purification.

Perovskite precursor solution

Normal-bandgap FAPbI₃-based perovskite (1.56 eV) solution (1.5 M, FA_{0.85}MA_{0.1}Cs_{0.05}Pb(I_{0.98}Br_{0.02})₃) was prepared by mixing 807 mg FAPbI₃ single-crystal powder, 138 mg PbI₂, 23.9 mg MAI, 19.5 mg CsI, 16.5 mg PbBr₂, and 12.3 mg MACl in the 1 ml mixed solvent of DMF and DMSO (volume ratio: 4:1) and stirred for 2 h. Wide-bandgap perovskite (1.68 eV) solution (1.3 M, Cs_{0.22}FA_{0.78}Pb(I_{0.85}Br_{0.15})₃) was prepared by mixing 641.8 mg FAPbI₃ single-crystal powder, 20.3 mg PbI₂, 74.3 mg CsI, and 107.3 mg PbBr₂ in mixed solvent of DMF and DMSO (volume ratio: 4:1) and stirred for 5 h. 5 mol% MACl (4.4 mg) and PbCl₂ (18.1 mg) with molar ratio were added to increase the crystallinity. Mini-modules perovskite: 1.2 M FA_{0.9}Cs_{0.1}PbI₃ was prepared by mixing 31.2 mg CsI, 55.3 mg PbI₂ and 683.6 mg FAPbI₃ single-crystal

powder in mixed solvent of DMF and DMSO (volume ratio: 4:1) and stirred for 2 h. 15 mol % MACl (12.2 mg) and 3 mol % PbCl₂ (10.0 mg) with molar ratio were added to increase the crystallinity. All the perovskite precursor solutions were filtered by 0.22 µm polytetrafluoroethylene (PTFE) filters before use.

Fabrication of perovskite solar cells

For the normal bandgap PSCs, the pre-patterned FTO glass substrates (2.5 × 2.5 cm, 10 Ω/sq.) were cleaned by ultrasonication in detergent and ultrapurewater (four times) for 30 min, respectively. The cleaned substrates were dried under N₂ flow and treated with ultraviolet ozone (UVO) for 15 min, then immediately transferred into an N₂ glove box (the oxygen content of less than 0.5 ppm and water content less than 0.1 ppm) in preparation for the deposition process. The hole transport layer was prepared with 4PADCB (0.5 mg mL⁻¹) dissolved in anhydrous ethanol and spin-coated onto the above-prepared FTO substrates, and then annealed at 100 °C for 10 min. The deposition of the perovskite layer was carried out after the substrates were cooled to room temperature. The perovskite layer was deposited by one-step spin-coating filtered precursor solution at 1,000 rpm for 10 s, and 4,000 rpm for 40 s, where 300 µL ethyl acetate was quickly dropped onto the center of the spinning substrate at 10 s prior to the end. The film was immediately annealed at 100 °C for 30 min. After cooled down to room temperature, the ligand engineering strategy was executed. Specifically, 1 mg mL⁻¹ TFEA/TFBA/TFPA/TFPmA solution (in IPA solvent) were drop-cast on the spinning perovskite film at 3000 rpm for 30 s without further annealing. Next, C₆₀ (20 nm) and BCP (5 nm) were thermally evaporated on the perovskite films under a high vacuum of <1 × 10⁻⁴ Pa. Finally, Ag electrodes (120 nm) were thermally evaporated under a vacuum of <5 × 10⁻⁴ Pa.

For the wide-bandgap PSCs, the substrate preparation process is the same as above. After that, Cs_{0.22}FA_{0.78}Pb(I_{0.85}Br_{0.15})₃ perovskite precursor solution was spin-coated on the FTO/4PADCB substrates at 4,000 rpm for 40 s. At the last 10 s, 200 µL ethyl acetate was dropped as the antisolvent. The film was immediately annealed at 100 °C for 50 min. After cooled down to room temperature, 1 mg mL⁻¹ TFPmA solution was drop-

cast on the spinning perovskite film surface at 3000 rpm for 30 s without further annealing. Finally, C₆₀ (20 nm)/BCP (5 nm)/Ag (120 nm) were sequentially deposited on top of perovskite by thermal evaporation.

Fabrication of perovskite solar modules

The fabrication of perovskite mini-modules began with the P1 laser (nanosecond green laser, scan speed = 1.0 m s⁻¹, laser power = 1.5 W, and repetition rate = 350 kHz.) scribing process to isolate the patterned FTO film. The 6×6 cm² FTO substrates were sequentially cleaned with detergent and ultrapurewater (four times) for 30 min, respectively. Then, the cleaned substrates were dried under N₂ flow and treated with UVO for 15 min before use. A NiO_x hole transport layer was then deposited by spin-coating a freshly prepared precursor solution of nickel acetylacetonate dihydrate (0.1 mmol mL⁻¹) dissolved in ethanol and ethanolamine (1:0.005, v/v), followed by thermal annealing in air. After cooling, a 1.5 mg mL⁻¹ PTAA solution in chlorobenzene was spin-coated and dried. The perovskite photoactive layer was subsequently deposited by spin-coating precursor solution, followed by thermal annealing to form a dense and uniform crystalline film. After that, the interface TFPmA layers were drop-cast at spinning perovskite layers with 3000 rpm for 30 s without further annealing. All wet processing steps are conducted in a Class 1000 cleanroom at 23~25 °C with relative humidity maintained below 45%.

The perovskite-coated substrates were then transferred to a vacuum deposition system for sequential thermal evaporation of 20 nm of C₆₀ and 5 nm of BCP. This was followed by the deposition of a 7 nm SnO₂ layer using spatial atomic layer deposition (ALD) at 90 °C, with tetrakis(dimethylamino)tin and water as precursors. The P2 laser (picosecond green laser, scan speed = 0.5 m s⁻¹, laser power = 0.2 W, and repetition rate = 200 kHz.) scribing process was subsequently performed in ambient atmosphere to selectively remove all functional layers above the FTO. A 150 nm Ag back electrode was then deposited by thermal evaporation, followed by P3 laser (picosecond green laser, scan speed = 0.5 m s⁻¹, laser power = 0.15 W, and repetition rate = 200 kHz.) scribing to define the sub-cell interconnections. Conductive copper tapes with

Sn95/Pb5 coating (140 mm in length) are applied to connect the electrode terminals. For encapsulation, a 10×10 cm² tempered glass sheet is used as the bottom substrate. A butyl rubber sealant was applied along the edges, and a PO-based encapsulant film is placed at the center. The 6×6 cm² perovskite mini-module was then aligned and placed on top, with the conductive tape fixed to the surrounding butyl sealant. A second layer of butyl rubber was applied to cover the conductive tape, followed by placement of a PO film over the back side of the device. A top 10×10 cm² tempered glass sheet is then added to complete the stack. The entire assembly was laminated under vacuum at 90 kPa and 120 °C for 15 min. After lamination, the module was ready for *J-V* testing.

Film characterization

SEM (SU-8020, Hitachi FE-SEM) was used to characterize the surface morphology and cross-sectional view of the films. AFM and KPFM images were obtained from the Bruker dimension icon instrument. The contact potential difference (CPD) measured by KPFM is related to the work function difference between the tip and the sample according to: $CPD = (\Phi_{tip} - \Phi_{sample})/e$, where Φ_{tip} and Φ_{sample} are the work functions of the tip and sample, respectively, and e is the elementary charge. Due to the uncertainty in the absolute tip work function, KPFM is used here to evaluate the relative surface potential distribution and spatial heterogeneity. Statistical parameters, including the mean absolute deviation (Ra) and standard deviation (σ), are extracted from pixel-by-pixel CPD data. XRD (2700BH, Dandong Haoyuan) patterns were measured with diffractometer with Cu K α ($\lambda = 1.54 \text{ \AA}$) radiation. UV visible spectrophotometer (Hitachi-UH4150) were performed to get the UV-Vis spectra of perovskite films and the wavelength detection range was set to 400 to 900 nm. PL and TRPL spectra were measured with a PicoQuant FluoQuant 300 and the laser with a filter wavelength of 510 nm. *In-situ* PL spectra were performed with a short-wavelength laser diode (405 nm), which aims to detect the surface region of films to minimize interference from the bulk material, coupled with a visible-range spectrometer under a dark environment at 25°C, 15-20% humidity. Moreover, the initial state of all perovskite films measured was identical. UPS and XPS spectra were obtained a photoelectron spectrometer

(ESCALAB250Xi, Thermo Fisher Scientific). Unless otherwise specified, all characterizations were performed on FAPbI₃-based perovskite (FA_{0.85}MA_{0.1}CS_{0.05}Pb(I_{0.98}Br_{0.02})₃) films fabricated following our standard one-step spin-coating protocol. ¹H-NMR spectra were measured by JNM-ECZ400R/S1 (JEOL, Japan) apparatus. Fourier-transform infrared (FTIR) spectra were collected using a Bruker Tensor II spectrometer equipped with KBr beam splitter.

Solar cell characterization

J-V and *I-T* curves were measured with source meter (Keithley 2450) and solar simulator (SAN-EIELECTRIC DM-50S1) under AM1.5G, which was adjusted using standard silicon cell. The voltage range for forward and reverse scans were 1.4~−0.1 V with a step of 0.02 V and a delay time of 20 ms. The contact area of the cell is 0.12 cm², and the illumination area is 0.074 cm². All devices were tested at room temperature (ca. 25 °C) in N₂ glove box. The EQE curves were performed on a QE-R system and the light source is a 300-W Xe lamp (Enli, QE-R). The EIS curves were measured with an electrochemical workstation (Modulab XM, USA). The M-S measurements were performed on an electrochemical workstation (Modulab XM, USA) and the frequency was set to 10 kHz and the scan voltage range was 0~1.5 V. The device structure was FTO/4PADCB/Perovskite/ligands/C₆₀/BCP/Ag. The built-in potential (*V*_{bi}) was obtained from the linear fitting derived from Equation: $C_{dl} = \sqrt{\frac{q\epsilon\epsilon_0 N}{2(V_{bi}-V)}}$, where *q* is the elementary charge, ϵ is the relative dielectric constant of the perovskite material, ϵ_0 is the vacuum permittivity, *N* is the carrier density.

Outdoor real-time field testing of perovskite solar modules

The outdoor stability measurements were conducted in Hangzhou of China, from February 2025 to October 2025 for 258 days for the TFPmA-treated module and from October to November 2025 for 27 days for the control module. All modules were subjected to continuous operation under daylight illumination and heat cycles from day to night. The temperature sensor (type: PT100) is placed on the back of the modules, which can reach a peak temperature of 68.9°C during the daytime and a minimum of

about -11.5°C at night. On sunny days, the light intensity reaches a maximum of 1.39 sun. All data is monitored in real-time and recorded in a computer.

Data availability

The data that support the findings of this study are available within the article and its Supplementary Information/Source Data file. Source data are provided with this paper.

Fundings

This work was supported by National Natural Science Foundation of China (61974085, K.Z.), Key Research and Development Program of Shaanxi (2025CY-YBXM-170, K.Z.), the 111 Project (B21005, S.L.), the National 1000 Talents Plan program (1110010341, S.L.). Young Scientist Initiative Project of School of Materials Science and Engineering at Shaanxi Normal University (2024YSIP-MSE-SNNU001). F.G. is a Wallenberg Scholar and acknowledges the support from the Swedish Energy Agency (P2022-00756 and P2025-03049, F.G.) and the Swedish Government Strategic Research Area in Materials Science on Advanced Functional Materials at Linköping University (Faculty Grant SFO-Mat-LiU No. 2009-00971).

Acknowledgements

We thank the staff of the BL17B beamline (<https://cstr.cn/31129.02.NFPS.BL17B>) at the National Facility for Protein Science in Shanghai (NFPS, <https://cstr.cn/31129.02.NFPS>), Shanghai Advanced Research Institute, Chinese Academy of Sciences, for their technical support in GIWAXS data collection and analysis. We also thank Dr. Ye Yuan and Dr. Genghua Yan (Jülich) for the useful discussion related to the quantifying efficiency limitations of devices. All authors discussed the results and commented on the manuscript.

Author contributions

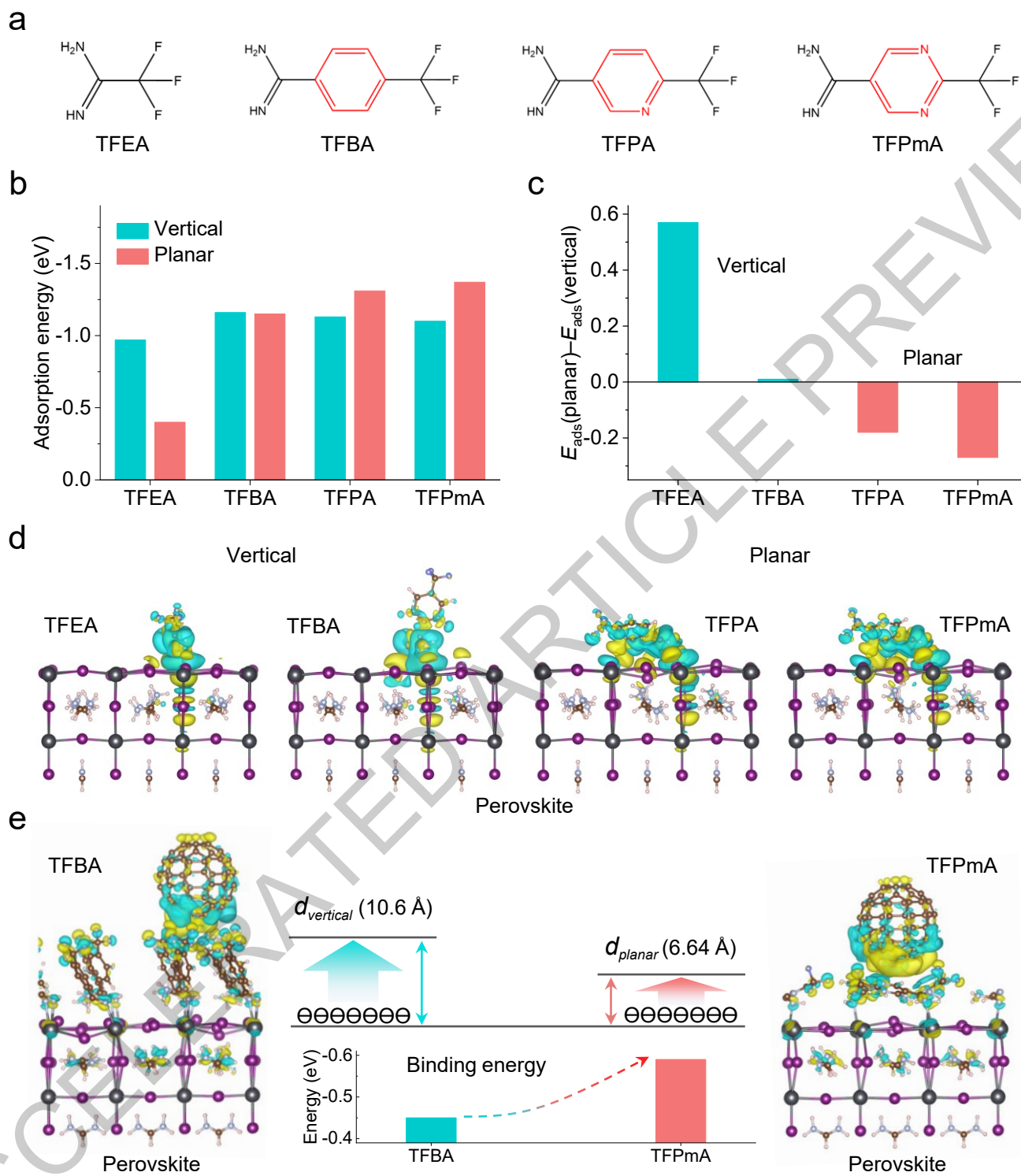
K.Z. conceived the concept and supervised the project. S.L., F.G. and K.Z. guided the research. T.Y. performed most of experimental characterization and data analysis. E.Z., Y.Y. and C.M. assisted with PL mapping procedure, C–V, FTIR and in situ PL measurement. N.W., X.C., C.T., T.N. and T.X. assisted with device fabrication. Z.Z., H.L., T.N. and N.X. participated in data analysis. X.C., S.W. and Z.D. helped with the GIWAXS measurement. H.W. helped with the nano IR measurement. L.Z., N.G. and B.Y. helped with the outdoor field-testing.

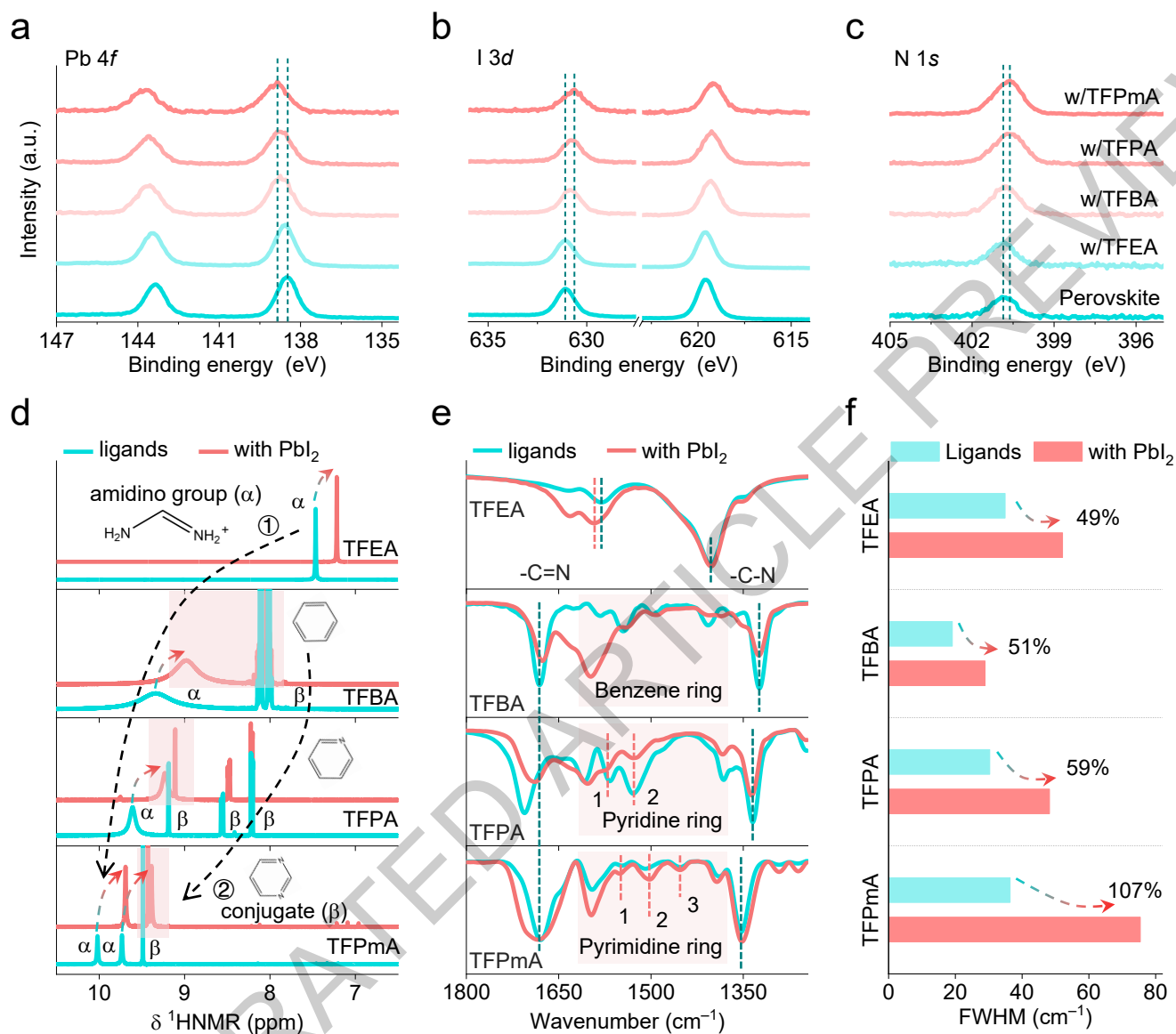
Competing interests

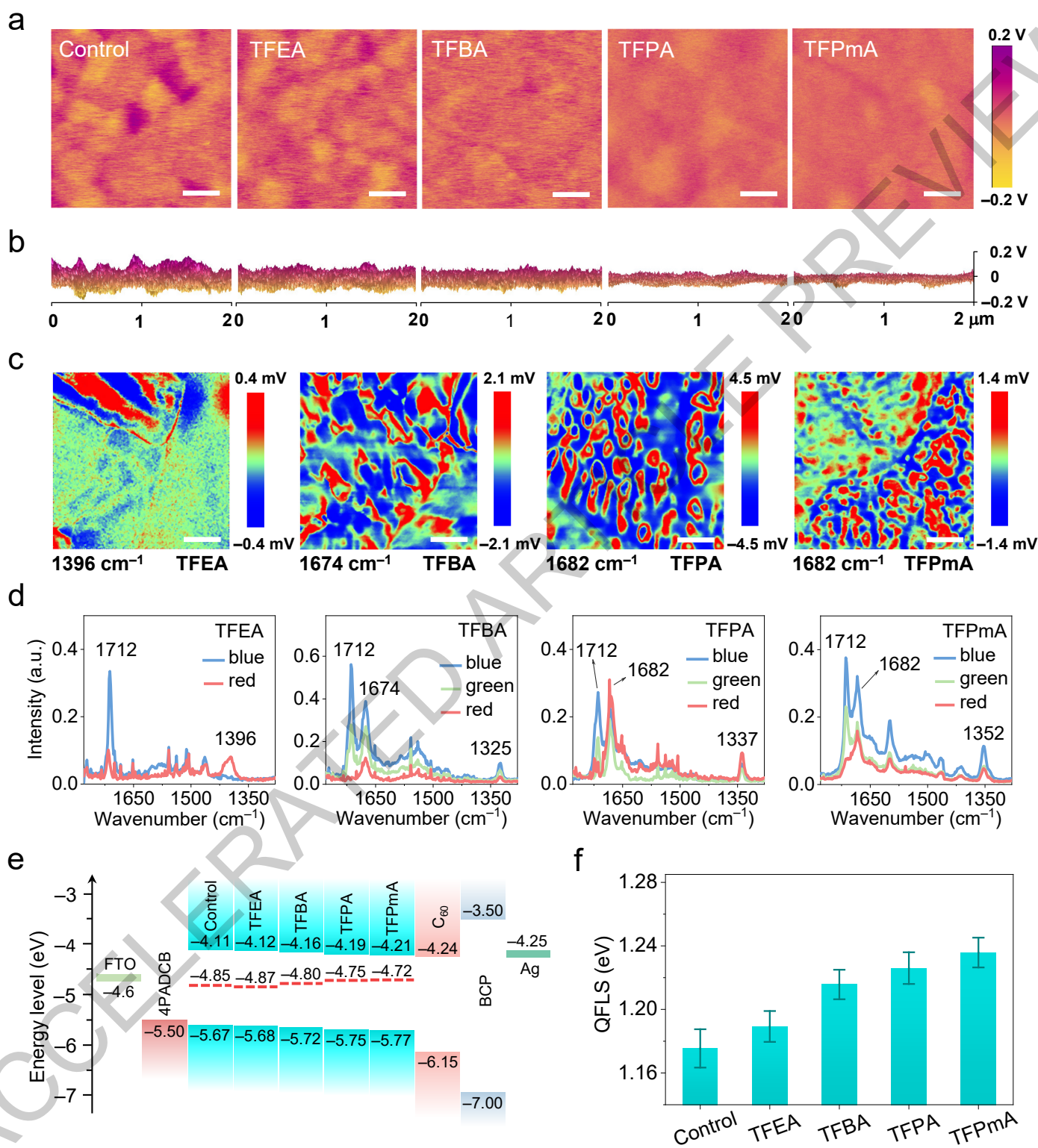
The authors declare no competing interests.

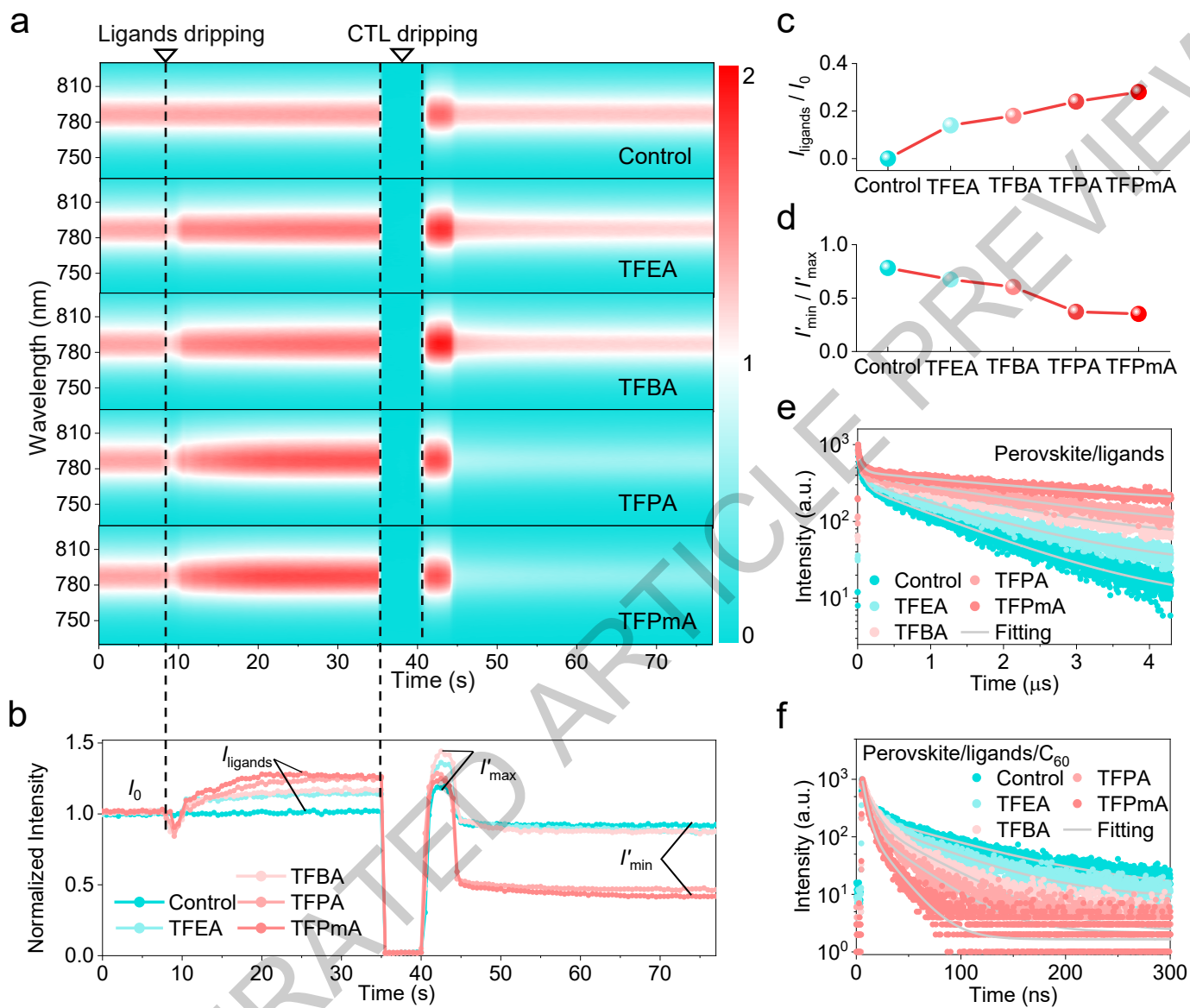
Additional information

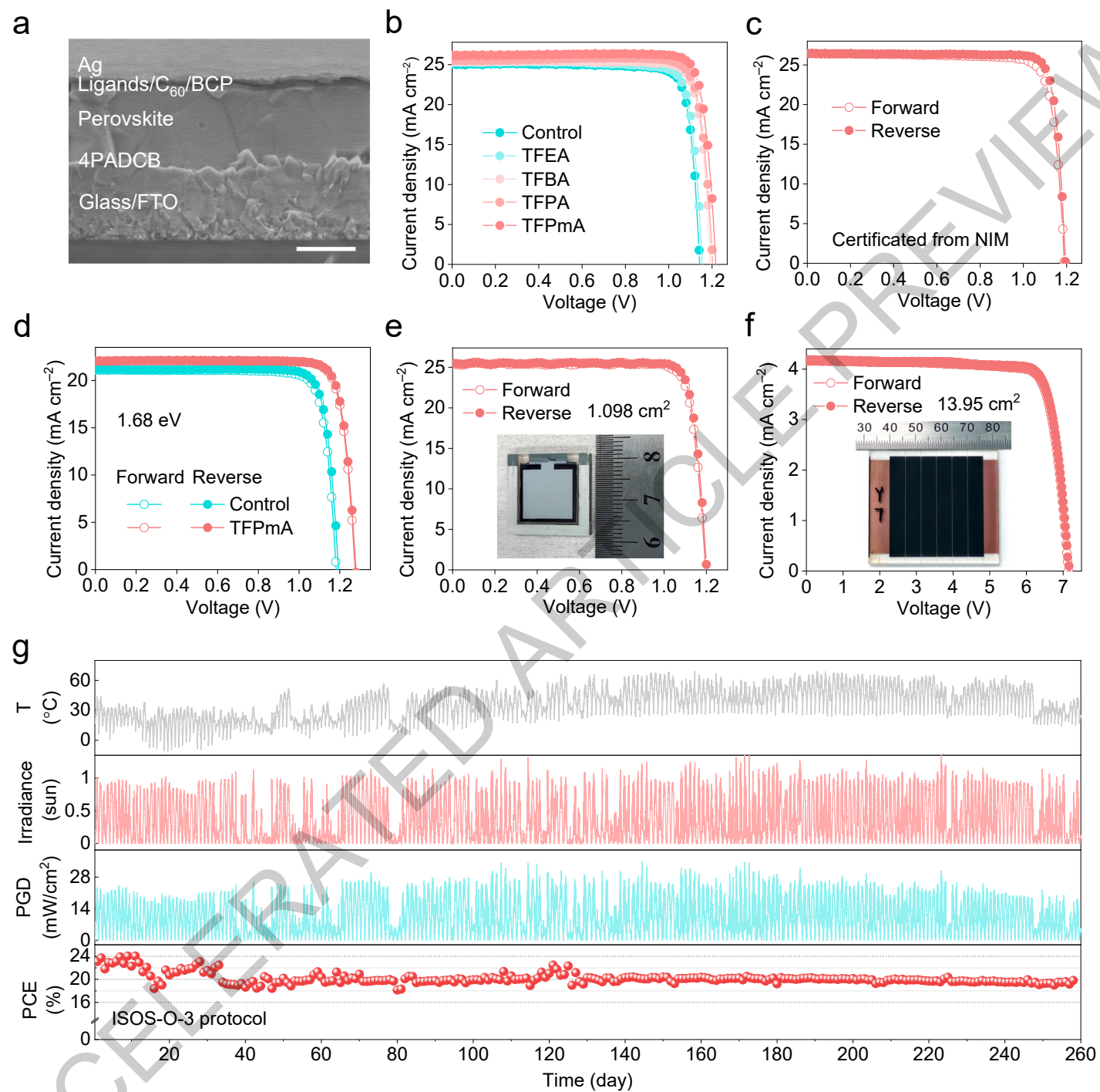
Supplementary Information The online version contains supplementary material available at xxxxx.











Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on Nature Research policies, including our [data availability policy](#), see [Authors & Referees](#).

► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

see Methods, 1.6 Solar cell characterization. The contact area of the cell is 0.12 cm², and the illumination area is 0.074 cm²

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

The area of our mask was certificated by the National Institute of Metrology (NIM).

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

The J-V plots measured along both scan direction, as shown in Figure 5c and Supplementary Fig. 37.

Voltage scan conditions

☒ Yes
☐ No

The voltage range for forward and reverse scans were 1.4~–0.1 V with a step of 0.02 V and a delay time of 20 ms. See Methods, 1.6 Solar cell characterization

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

All devices were tested at room temperature (ca. 25 °C) in N₂ glove box. See Methods, 1.6 Solar cell characterization

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☒ Yes
☐ No

Figure 5g, the real-time field testing (ISOS-O-3 protocol) of TFPmA-treated module under actual out-door operating conditions for 258 days. Figure S46, thermal aging test (ISOS-D-2 protocol) of the control and TFPmA treated devices at 85 °C. Figure S47, MPP tracking test (ISOS-L-1 protocol) of the control and TFPmA treated devices under light soaking.

Explain why this information is not reported/not relevant.

Stability of the J-V characteristic

☒ Yes
☐ No

The stabilized power output of solar cell is shown in Figure S38, and the stabilized PCE of 26.85% at bias voltage 1.07V.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

see Figure S37 and Table S7. The forward and reverse scan efficiencies of the device are 27.58% and 27.01%, respectively, with a hysteresis factor of 0.021.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

The forward and reverse scan data are shown in Figure S37 and Table S7.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

The external quantum efficiency (EQE) measurements was shown in Figure S42.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The external quantum efficiency (EQE) integrated currents agree with J-V measurements (tolerance variation < 2.1%). <i>Explain why this information is not reported/not relevant.</i>
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	<i>Provide a description of the measurement conditions.</i> This is not a tandem solar cell

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	J-V and I-T curves were measured with source Meter (Keithley 2450) and solar simulator (SAN-EIELECTRIC DM-50S1) under AM1.5G. <i>Explain why this information is not reported/not relevant.</i>
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The calibrated Si-reference cell were certificated by NREL. <i>Explain why this information is not reported/not relevant.</i>
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	See Figure S41. The relative spectral response (SR) curves measured at National Institute of Metrology, China (NIM, China) and the corresponding data table. <i>Explain why this information is not reported/not relevant.</i>

6. Mask/aperture

Size of the mask/aperture used during testing	☒ Yes ☐ No	The size of mask during certification is 0.0742 and 0.0718 cm ² , respectively, as shown in Figure S39, S45. <i>Explain why this information is not reported/not relevant.</i>
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	<i>Report the difference in the short-circuit current density values measured with the mask and aperture area.</i> We did not do the current variation test with different mask areas.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Some cells were sent to National Institute of Metrology (NIM, China) and National PV Industry Measurement and Testing Center (NPVM, China) for certification. <i>Explain why this information is not reported/not relevant.</i>
A copy of any certificate(s)	☒ Yes ☐ No	The related test report are shown in Figure S38-41, S44,45. <i>Explain why this information is not reported/not relevant.</i>

8. Statistics

Number of solar cells tested	☒ Yes ☐ No	The statistics of the parameters of solar cells includes 20 individual devices are shown in Figure S43. <i>Explain why this information is not reported/not relevant.</i>
Statistical analysis of the device performance	☒ Yes ☐ No	Statistical parameters of 20 devices without and with ligands treatment, as shown in Figure S43 and Table S8 <i>Explain why this information is not reported/not relevant.</i>

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Figure 5g, the real-time field testing (ISOS-O-3 protocol) of TFPmA-treated module under actual out-door operating conditions for 258 days. Figure S46, thermal aging test (ISOS-D-2 protocol) of the control and TFPmA treated devices at 85 °C. Figure S47, MPP tracking test (ISOS-L-1 protocol) of the control and TFPmA treated devices under light soaking. <i>Explain why this information is not reported/not relevant.</i>
--	--	--

